

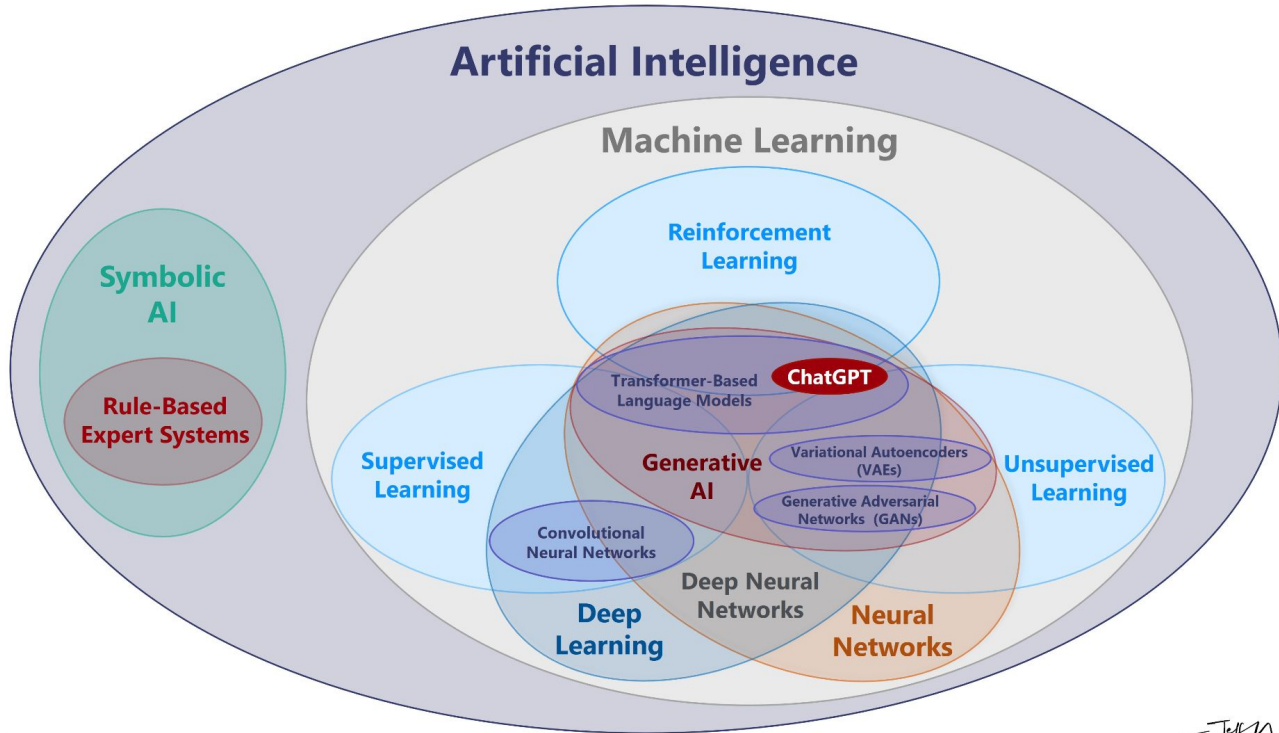
Data, ML, and AI – What are they?

CU Boulder AI Club Workshop Series

Tate Middleton - 10/15/25



Outlook



Today:

- What is data?
 - Types of data
- How do we use data: ML and AI
 - ML: Learning from data
 - AI: A machine doing tasks with “human-like intelligence”

Jeff M.

What is data?

- Everything that is recorded is data!
 - Financial records
 - Images
 - Audio
 - Engagement (likes, comments, views)
 - *Your actions online (clicks, views, purchases, personal info)*
- Artificial vs. natural data
- We can extract meaningful info from these piles of data

Types of Data

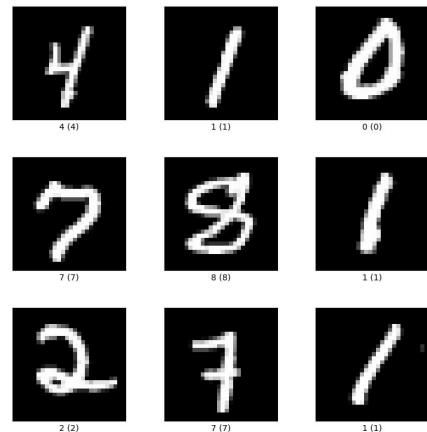
- Structured (think *labeled* or *rows and columns*):
 - Tables
 - Spreadsheets
- Unstructured (can be more difficult to use):
 - Images
 - Bodies of text

B13 $\times$ $\checkmark$ f_x $=SUM(B3:B12)$

	A	B	C	D	E
1		Sales in Each Quarter			
2	Product Name	Jan'2018	April'2018	July'2018	October'2018
3	ABC Mutton	\$ 2,667.60	\$ 4,013.10	\$ 4,836.00	\$ 6,087.90
4	Crab Meat	\$ 1,768.41	\$ 1,978.00	\$ 4,412.32	\$ 1,656.00
5	Camembert Pierrot	\$ 3,182.40	\$ 4,683.50	\$ 9,579.50	\$ 3,060.00
6	Ipoh Coffee	\$ 1,398.40	\$ 4,496.50	\$ 1,196.00	\$ 3,979.00
7	Hot Pepper Sauce	\$ 1,347.36	\$ 2,750.69	\$ 1,375.62	\$ 3,899.51
8	Hot Spiced Okra	\$ 1,509.60	\$ 530.40	\$ 68.00	\$ 850.00
9	Mozzarella di Giovanni	\$ 1,390.00	\$ 4,488.20	\$ 3,027.60	\$ 2,697.00
10	Sir Rodney's Scones	\$ 1,462.00	\$ 644.00	\$ 1,733.00	\$ 1,434.00
11	Steeleye Stout	\$ 1,310.40	\$ 1,368.00	\$ 1,323.00	\$ 1,273.50
12	Veggie-spread	\$ 3,202.87	\$ 263.40	\$ 842.88	\$ 2,590.10
13	Grand Total	\$ 19,239.04			
14					

[2]

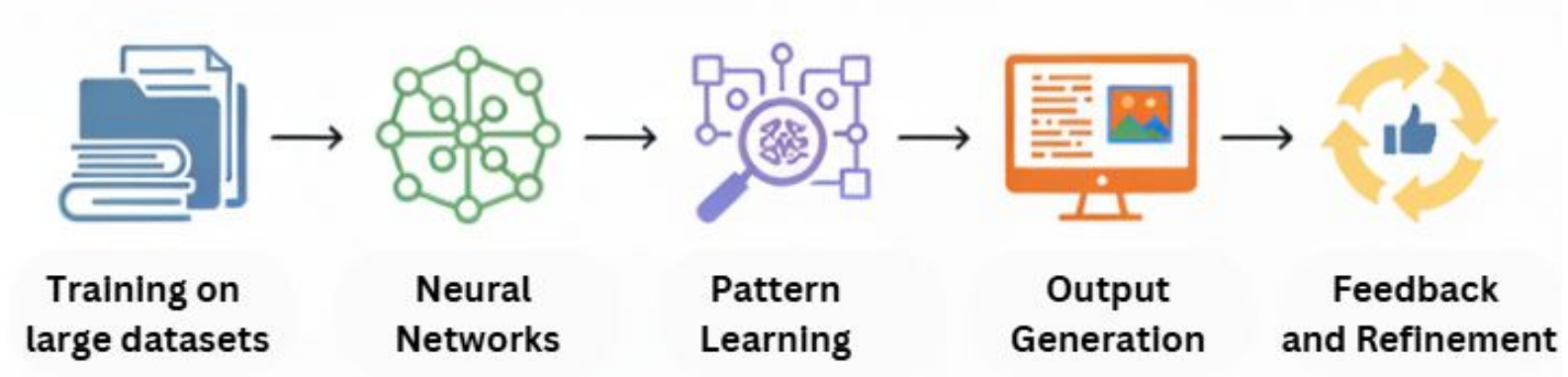
VS



“MNIST” Dataset

Data + Artificial Intelligence

- AI is made for very complex tasks $\Rightarrow$ Needs **lots of data** to know how
 - AlphaGo, Google's Go playing AI - played **30 million games** against itself
 - ChatGPT: Trained on **9.175 trillion words**
 - Tesla self-driving: **100 years** worth of driving videos



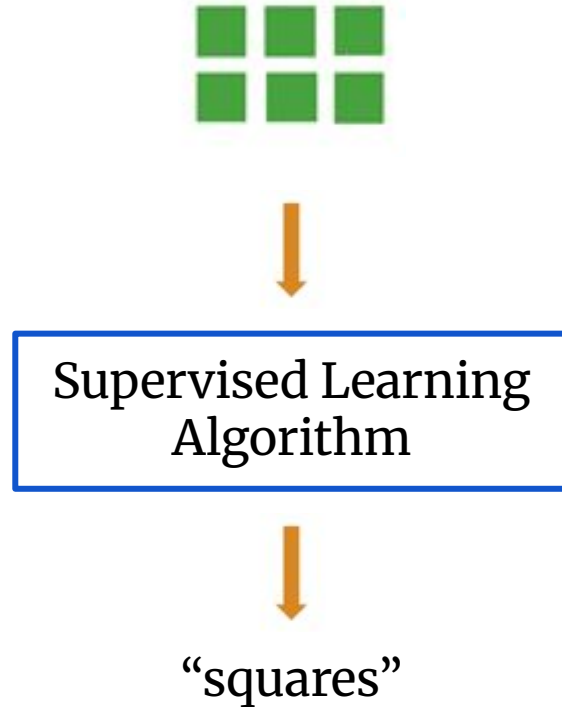

Or other algorithms!

Artificial Intelligence + Machine Learning

- To create AI, we employ multiple types of ML in one system
- Main types of ML:
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning

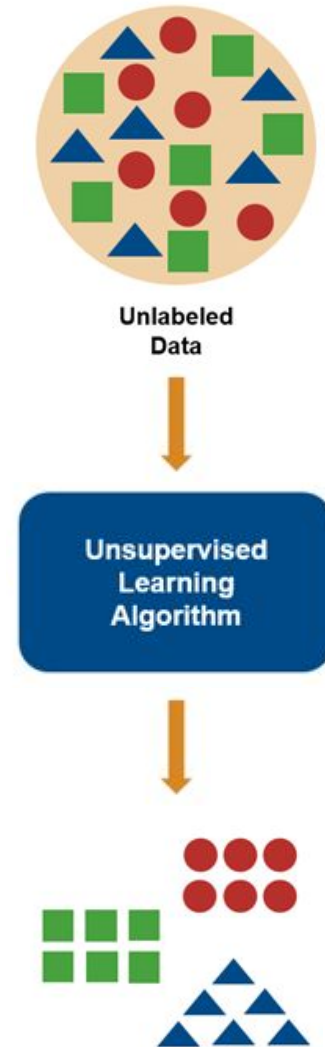
Supervised Learning

- Data Type: Labeled data (input-output pairs)
- Goal: To learn a mapping from inputs to outputs so the model can predict the output for unseen inputs
- Feedback: Feedback provided by the correct labels
- Typical Problems:
 - **Classification**: Predicting a categorical output (“Is this email spam or not spam?”)
 - **Regression**: Predicting a continuous value (“What is the price of this house?”)
- Algorithms: Support vector machines, linear regression



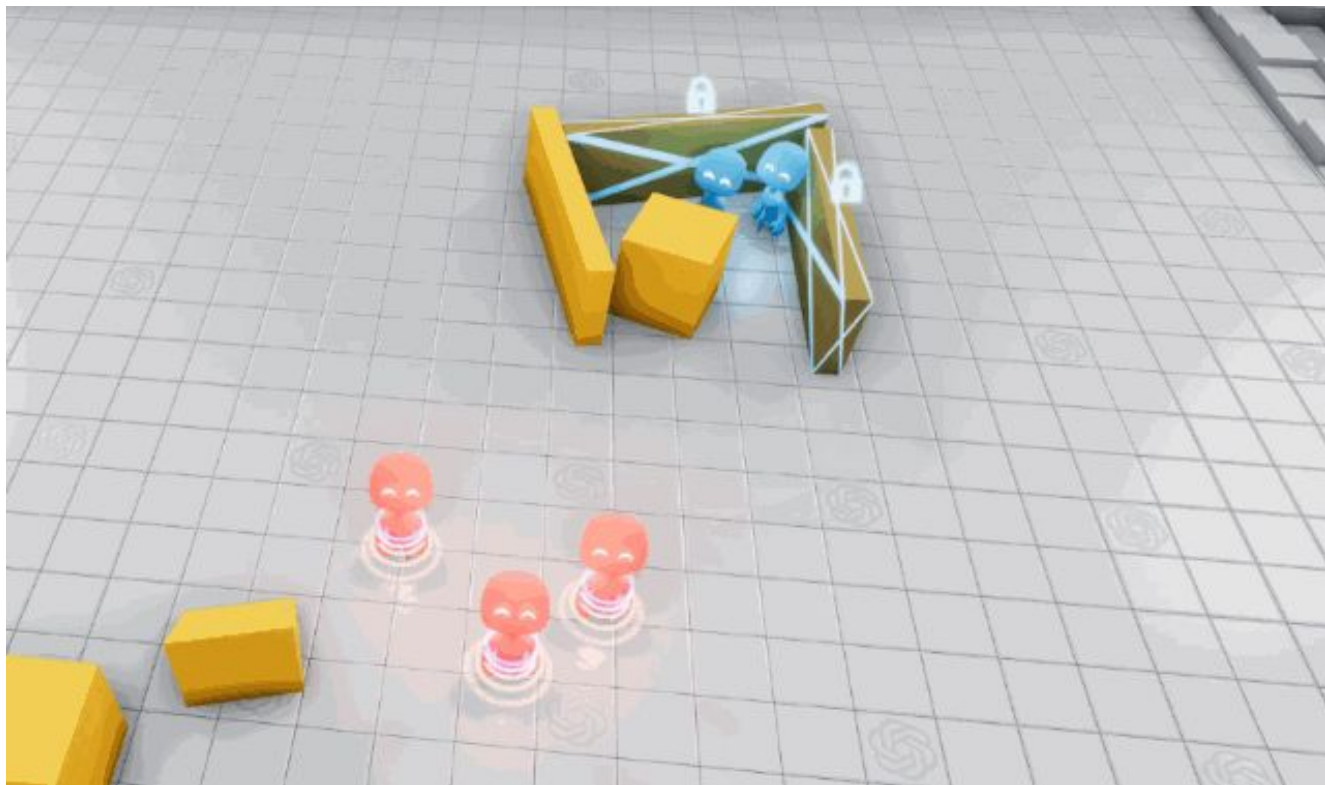
Unsupervised Learning

- Data Type: Unlabeled data (inputs only)
- Goal: To discover inherent structure and patterns in the data
- Typical Problems:
 - **Clustering**: Grouping similar data points together (flower types by color)
 - **Association**: Discovering rules to describe relationships between variables ("People who buy milk often buy bread")
- Algorithms: K-Nearest Neighbors, Autoencoders



Reinforcement Learning

- Data Type: Data is generated through interaction with the environment
- Goal: To learn an optimal sequence of actions (a policy) that maximizes the total, long-term reward
- Feedback: Received as a reward or penalty from the environment
- Typical Problems:
 - **Sequential decision-making** (training a self-driving car or a robot)
 - **Game playing** (AlphaGo)



Recap

- Data is everywhere and gives life to AI and ML
- ML is algorithms that learn from data
- AI uses ML + other training protocols to do something as good, or better than a human
- Next time:
 - Neural networks (the muscle of AI)
 - Using advanced AI tools
- Offweek practice:
https://colab.research.google.com/drive/12bgs1AeJM7CiH1w_c2DDcWhCSe1McyYO?usp=sharing

More Resources

- 3blue1brown: <https://www.3blue1brown.com/>
- Feedback:
<https://forms.gle/4122RR7pzJcwrJC79>
- Other resources:
<https://docs.google.com/document/d/1YaS8a6d095RFVoTop9ZsP-y1AFcW-GlE9uj7Z0K3EDo/edit?usp=sharing>



Citations

[1]<https://www.jeffwinterinsights.com/s/Artificial-Intelligence-Topology-v30-c58c.jpg>

[2]https://res.cloudinary.com/monday-blogs/w_1748,h_1070,c_fit/fl_lossy,f_auto,q_auto/wp-blog/2022/02/1535-image2.png

[3]https://www.mathworks.com/discovery/unsupervised-learning/jcr_content/mainParsys/band/mainParsys/lockedsubnav_copy/mainParsys/columns/335ce30a-77fd-4c27-af75-93672733f56b/columns/0e1a10a0-b289-4f36-9928-f1add97647b8/image_2128876021_cop.adapt.full.medium.png/1760345171047.png